PathwayAI | Dutchess County Lyme Burden & Action Report

PathwayAI: Making the Hidden Patient Journey Visible—and Actionable

External review copy · Public evidence and support planning · Version 1.1-community-action

Prepared: 2026-11-01 · County surveillance: 2024 · Study costs: 2016 dollars

Newer official county releases have not been verified for this brief.

1. OBSERVED PUBLIC HEALTH DATA

Reported Lyme cases: 1,060 · 2024 · county FIPS 36027.

Reported rate: 355.5 per 100,000 residents. New York State: 109.8.

NYSDOH reports cases by residence, not all infections or where exposure occurred. CDC historical county files in the app cover 2019–2022; they are not combined with 2024 counts.

Published rates are reproduced; exact population denominators and their vintage have not been independently verified.

Healthcare resource context · HRSA AHRF 2024–2025 release · 2023 fields

Primary-care physicians: 233

Hospitals: 2

Hospital beds: 404

Existing public AHRF asset; 2023 county fields.

General healthcare resource context, not Lyme expertise, appointment availability, interpreter access or a verified shortage.

2. PUBLISHED RESEARCH ESTIMATES AND COUNTY ILLUSTRATION

1,060 reported cases × USD 2,032 study mean = USD 2,153,920 (2016 dollars).

Hook et al. (published 2022): 613 participants in societal analysis; reported cases recruited in four high-incidence states during 2014–2016.

Societal: medical, nonmedical and productivity costs; components are already included.

Applying this study mean to reported county cases is an assumption; the result is not measured local spending.

No adjustment for underreporting or local case mix. No causal savings or disability probabilities are inferred.

This illustrates scale, not today’s local costs. Hook excluded non-English speakers and nonacute disease; the study does not represent every local household.

Optional 2025 purchasing-power comparison · 11 available months

Study mean in 2025 purchasing power: USD 2,726 per participant. County illustration in 2025 purchasing power: USD 2,889,245.

General inflation only. 2016 mean CPI: 240.00716666666666666666666667; 2025 available-month mean: 321.943; factor: 1.34138911129737098128319 7793

2025 has 11 observed months; October is missing. No imputation. This is not a verified official annual-average restatement.

Inflation changes purchasing power, not treatment patterns or local representativeness.

https://fred.stlouisfed.org/series/CPIAUCNS

Source workbook SHA-256: 346ef77b0f6cc7bb59c42d27be2c802a5aefb3f518339191e9320a1cb105d28a

3. PATIENT SURVEY FINDINGS · NOT COUNTY ESTIMATES

Snapshot: 2026-10-03 · 33 respondents

Small, self-selected sample across locations; respondents self-reported a Lyme diagnosis.

The supplied summary raises questions about spending, time lost, repeated visits and diagnosis delays.

Not representative of the selected county or the U.S. population. Raw responses, question-specific denominators and cost/time reporting periods have not been verified here; no survey percentages or county totals are calculated.

Project-lead-supplied summary already documented in the app; not a new analysis of raw responses.

Only this aggregate context is included. Personal app entries are not included in community evidence.

What we still need to understand

- Lyme-attributable county disability and duration
- Continuing costs beyond the study follow-up
- Local unmet care and support needs
- Experiences of people excluded or underrepresented in the source study

4. FROM LYME DISEASE BURDEN TO COMMUNITY ACTION

Suggested actions for local review, not clinical advice or adopted programs.

Prevention education and tick-bite awareness: Share established prevention information through trusted local channels.

Timely access to healthcare: Help residents find care and ask about appointments, costs, transport and interpreters.

Financial and caregiving burden: Recognize spending, missed work, unpaid care and support needs beyond medical bills.

Language access and community resources: Make information easier to understand and connect residents with existing local support.

Planning audience: Public health departments

- Share tick-prevention information through local events and trusted community channels.
- Review county surveillance alongside service availability; ask residents which barriers need attention.

- Bring residents and potential partners together to choose one support priority and agree how to follow up.

5. A SUPPORT QUESTION TO DISCUSS

Care without insurance and medical bills

Planning question: Could existing services meet reported uninsured-care needs, or would an additional funded response be needed?

Possible response: Review health-center sliding-fee access, coverage enrollment help and hospital financial assistance with patients; identify staffing, appointment capacity, costs and a possible funding source before proposing new services.

Who could help: Patients, health centers, hospital financial-assistance staff, enrollment assisters, county planners and potential funders

These are proposed discussion topics, not findings about local demand or commitments from partners.

6. WHAT TO CHECK BEFORE STARTING A PILOT

- Confirm local demand with residents and caregivers, including assisted/offline participation.
- Verify current services, eligibility, interpreter access and appointment capacity.
- Agree who is responsible and estimate staff time, program costs and a funding source.
- Define consent, minimal data collection, access, retention and deletion before any collection.

7. CHECK WHETHER PEOPLE RECEIVED SUPPORT

Choose a common follow-up period, such as 30 days. Record the starting situation and support received. Keep unanswered questions marked as unknown.

For this priority, check: Care appointments reached; enrollment assistance completed; financial-assistance decisions; remaining bills and unmet needs over a common period

Use findings to improve support. Before/after changes alone do not establish that the tool caused improvements or saved money.

8. DATA SOURCES AND METHODS

Reported cases, county of residence · 2024

https://www.health.ny.gov/statistics/diseases/communicable/2024/docs/cases.pdf

Previously checked supplied official PDF, page 5.

Reported rates per 100,000 · 2024

https://www.health.ny.gov/statistics/diseases/communicable/2024/docs/rates.pdf

Previously checked supplied official PDF, page 5.

CDC surveillance interpretation; historical county file in app · 2019–2022 county file; separate from NYSDOH 2024

https://www.cdc.gov/lyme/data-research/facts-stats/index.html

Background surveillance context, not an additional 2024 county estimate.

General healthcare resources · 2023 fields; AHRF 2024–2025 release

https://data.hrsa.gov/topics/health-workforce/nchwa/ahrf

Existing public AHRF asset; 2023 county fields.

Societal study mean · 2016 dollars; recruitment 2014–2016

https://wwwnc.cdc.gov/eid/article/28/6/21-1335_article

Hook et al., published 2022; participant recruitment 2014–2016; study costs expressed in 2016 dollars.

Public evidence and proposed planning actions only. No session stories, personal records or synthetic patient cards are included.

Discussion brief, not a county demand assessment, benefits decision, funded service or adopted county plan.